P8X Instruction Set -- Quick Reference (rev D)

Opcodes, mnemonics and cycle counts generated live from genucode.py (the microcode source of truth) -- cannot drift from the hardware.

Operands: #imm immediate | addr 16-bit absolute | (Pn) ptr indirect | (Pn)+ post-increment | (no operand) implied. **By**=bytes, **Cy**=cycles (incl. fetch). **Flags (Fl):** C carry (active-high: ADD carry-out / SUB,CMP no-borrow A>=B), Z zero, N negative (bit7), V overflow. '-' = none. Signed branches test N^V / Z. **Memory (rev E):** $0000-1FFF ROM | $2000-FEFF RAM | $FF00-FFFF I/O. P0=PC, P3=stack (empty-descending). JZ/JNZ/JC are aliases of BZ/BNZ/BCP.

Op	Mnemonic	By	Cy	Fl	Description
System					
$00	NOP	1	2	-	No operation.
$01	HLT	1	2	-	Halt clock; resume only by reset.
$72	CLC	1	2	C	C:=0.
$73	SEC	1	2	C	C:=1.
Interrupts (rev C)					
$02	EI	1	2	-	Enable maskable interrupts (IE:=1).
$03	DI	1	2	-	Disable maskable interrupts (IE:=0).
$04	RTI	1	10	-	Return from interrupt: pop flags then PC; re-enable IE.
$08	IRQ	1	10	-	SW interrupt: push PC+flags, vector to $0808.
Load / store					
$10	LDA #imm	2	2	ZN	A:=immediate.
$11	LDB #imm	2	2	ZN	B:=immediate.
$12	LDA addr	3	6	ZN	A:=byte at addr (absolute).
$13	LDB addr	3	6	ZN	B:=byte at addr (absolute).
$14	STA addr	3	6	-	byte at addr:=A (absolute).
$15	LDA (P1)+	1	2	ZN	A:=[P1], P1++.
$16	LDA (P2)+	1	2	ZN	A:=[P2], P2++.
$17	LDA (P3)+	1	2	ZN	A:=[P3], P3++.
$19	STA (P1)+	1	2	-	[P1]:=A, P1++.
$1A	STA (P2)+	1	2	-	[P2]:=A, P2++.
$1B	STA (P3)+	1	2	-	[P3]:=A, P3++.
$1D	STA (P1)	1	2	-	[P1]:=A.
$1E	STA (P2)	1	2	-	[P2]:=A.
$1F	STA (P3)	1	2	-	[P3]:=A.
$51	LDA (P1)	1	2	ZN	A:=[P1] (P1 kept).
$52	LDA (P2)	1	2	ZN	A:=[P2] (P2 kept).
$53	LDA (P3)	1	2	ZN	A:=[P3] (P3 kept).
ALU (A,B -> A)					
$20	ADD	1	2	CZN	A:=A+B.
$21	SUB	1	2	CZN	A:=A-B.
$22	AND	1	2	CZN	A:=A AND B.
$23	OR	1	2	CZN	A:=A OR B.
$24	XOR	1	2	CZN	A:=A XOR B.
$25	CMP	1	2	CZN	Flags from A-B; A unchanged.
$26	INC	1	2	CZN	A:=A+1 (B unused).
$27	DEC	1	2	CZN	A:=A-1 (B unused).
$28	SHL	1	2	CZN	A:=A<<1, 0->bit0, out->C.
$29	SHR	1	2	CZN	A:=A>>1, 0->bit7, out->C.
$2A	ROL	1	2	CZN	Rotate A left through carry.
$2B	ROR	1	2	CZN	Rotate A right through carry.
ALU with T (rev C; 2nd operand=T, B preserved)					
$80	ADDT	1	2	CZN	A:=A+T (B preserved).
$81	SUBT	1	2	CZN	A:=A-T (B preserved).
$82	ANDT	1	2	CZN	A:=A AND T (B preserved).
$83	ORT	1	2	CZN	A:=A OR T (B preserved).
$84	XORT	1	2	CZN	A:=A XOR T (B preserved).
$85	CMPT	1	2	CZN	Flags from A-T; A,B unchanged.
$86	LDT #imm	2	2	-	T:=immediate.
$87	LDT addr	3	6	-	T:=byte at addr (absolute).

Op	Mnemonic	By	Cy	Fl	Description
Stack					
$70	PHA	1	2	-	Push A onto P3 stack.
$71	PLA	1	3	ZN	Pop A from P3 stack.
16-bit memory (rev D)					
$74	PHW addr	3	10	-	Push 16-bit word at addr (lo then hi).
$75	PLW addr	3	12	-	Pop 16-bit word into addr.
$76	LPW1 addr	3	9	-	P1 := 16-bit word at addr.
$77	LPW2 addr	3	9	-	P2 := 16-bit word at addr.
$78	MOVW dst,src	5	13	-	16-bit mem->mem: word at src -> dst.
Control flow					
$40	JMP addr	3	5	-	P0(PC):=addr.
$41	JSR (P1)	1	9	-	Push return addr, P0:=P1.
$42	RTS	1	7	-	Pop return addr from P3 into P0.
$43	JSR addr	3	13	-	Push return addr, P0:=addr.
$48	BZ addr	3	5	-	Branch if Z=1. (JZ alias.)
$49	BNZ addr	3	5	-	Branch if Z=0. (JNZ alias.)
$4A	BCP addr	3	5	-	Branch if C=1 / A>=B unsigned. (JC alias.)
$4C	JNC addr	3	5	-	Branch if C=0 / A<B unsigned.
Signed branches (rev C; after CMP)					
$44	BLT addr	3	5	-	Branch if signed A<B (N^V=1). After CMP.
$45	BGE addr	3	5	-	Branch if signed A>=B (N^V=0). After CMP.
$46	BLE addr	3	5	-	Branch if signed A<=B ((N^V)|Z). After CMP.
$47	BGT addr	3	5	-	Branch if signed A>B. After CMP.
Pointer registers					
$31	LPL1 #imm	2	3	-	P1 low byte:=imm.
$32	LPL2 #imm	2	3	-	P2 low byte:=imm.
$33	LPL3 #imm	2	3	-	P3 low byte:=imm.
$35	LPH1 #imm	2	3	-	P1 high byte:=imm.
$36	LPH2 #imm	2	3	-	P2 high byte:=imm.
$37	LPH3 #imm	2	3	-	P3 high byte:=imm.
$54	INP1	1	2	-	P1:=P1+1.
$55	INP2	1	2	-	P2:=P2+1.
$56	INP3	1	2	-	P3:=P3+1.
$58	DEP1	1	2	-	P1:=P1-1.
$59	DEP2	1	2	-	P2:=P2-1.
$5A	DEP3	1	2	-	P3:=P3-1.
$5E	TAP1L	1	2	-	P1 low:=A.
$5F	TAP1H	1	2	-	P1 high:=A.
$60	TAP2L	1	2	-	P2 low:=A.
$61	TAP2H	1	2	-	P2 high:=A.
$62	TAP3L	1	2	-	P3 low:=A.
$63	TAP3H	1	2	-	P3 high:=A.
$68	TPA1L	1	2	ZN	A:=P1 low.
$69	TPA1H	1	2	ZN	A:=P1 high.
$6A	TPA2L	1	2	ZN	A:=P2 low.
$6B	TPA2H	1	2	ZN	A:=P2 high.
$6C	TPA3L	1	2	ZN	A:=P3 low.
$6D	TPA3H	1	2	ZN	A:=P3 high.